	7.9. METROLOGICAL TRACEABILITY	
AzurESky "ensure quality of test"	Document no: ASMF7.9	
	Compiled by:	**Sifiso Nyambi - Laboratory Manager**
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Metrological Traceability

Objective

To ensure that all certified values assigned to reference materials are metrologically traceable to the international system of units (SI units).

1. Requirements

- Certified values shall be traceable to the SI unit or other recognised accepted standard.
- Should SI traceability not be possible, the laboratory shall use other technically justifiable references such as consensus values, comparison with CRMs.
- The laboratory shall document and justify traceability route and maintain evidence to support it.

2. Establishing traceability route

Parameter	Unit	Required traceability to	Standard/reference
Calorific Value	MJ/kg	SI (Joules) via CRM (benzoic acid)	Benzoic acid tablet from supplier
Ash Content	%	SI via grams (g)	Calibrated balances traceable to SI
Moisture	%	SI- Mass difference	Traceable mass calibration
Volatile	%	SI- Mass and temperature	Calibrated oven and balance
Total Sulphur	%	SI via grams (g)	-

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3. Required elements of traceability system for the laboratory

A. Use of traceable standards and instruments
- Use CRM: e.g. benzoic acid tablets (bomb calorimetry)
- Calibrate instruments using services from Mass & Temperature metrology ISO/IEC 17025 accredited laboratory's: Balances, thermometers, ovens, bomb calorimetry.

B. Validation
- Ensure that all methods (ISO 1928, ISO 1171 etc) are validated in-house and that all standards are traceable.

C. Establishing the traceability chain
e.g. Benzoic acid CRM (with CoA stating SI traceability) → Calorimeter → Coal sample → Assigned value

4. Traceability statements

For e.g. :

- "The assigned value for Gross Calorific Value is traceable to the International System of Units (SI) through calibration with certified benzoic acid, supplied by an ISO 17034-accredited producer. All equipment used is calibrated by ISO/IEC 17025-accredited laboratories".
- "The certified value for Ash Content is traceable to the SI unit of mass (grams) through gravimetric determination using balances calibrated by ISO/IEC 17025-accredited laboratories. The method applied is based on ISO 1171, and the results are supported by the use of traceable mass standards and validated furnace temperature control".

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- "The certified value for Volatile Matter is traceable to the SI units of mass (grams) and temperature (Kelvin), using calibrated analytical balances and high-temperature ovens verified by ISO/IEC 17025-accredited laboratories. The determination method complies with ISO 562. Traceability is maintained via certified reference materials and controlled thermal decomposition conditions."
- “The certified value for Inherent Moisture is traceable to the SI unit of mass (grams) through mass loss determination using a balance calibrated by an ISO/IEC 17025-accredited laboratory. The method used is in accordance with ISO 589, and traceability is supported by certified mass standards and validated drying procedures with controlled temperature and time settings."
- "The certified value for Total Sulphur is traceable to the SI unit of mass fraction (mg/kg) through calibration with certified reference materials supplied by ISO 17034-accredited producers. Measurement instrumentation is calibrated and verified by ISO/IEC 17025-accredited laboratories, and determinations are performed using validated combustion or IR detection methods in accordance with ISO 19579 (or ASTM D4239)."